Digital Twin for Well-to-Surface Optimization of Cyclic Steam Stimulation (CSS) and Sucker Rod Pump (SRP) Operations for Heavy Oil Wells of Baghewala Field.

Stage 1: Steam Heat Delivery (Injection / Huff Phase)

Physical Principle

High-pressure boiler steam is injected downhole to heat the cold formation and crude oil. Steam transfers energy in two forms: sensible heat (as liquid cools) and latent heat (h_{fg}, released when vapor condenses to liquid).

Mathematical Formulation

1. **Total Injected Thermal Energy (Q_{inj})**:

$$Q_{\text{inj}} = m_s\left[x_{\text{qual}}h_{\text{fg}} + C_w(T_{\text{steam}} - T_{\text{res}})\right]$$

   - m_s: Total mass of steam injected (kg).
   - x_{qual}: Steam quality, dimensionless fraction of vapor ($0.0 \leq x_{\text{qual}} \leq 1.0$).
   - h_{fg}: Latent heat of vaporization (J/kg).
   - C_w: Specific heat capacity of liquid water (J/kg ·° C).
   - T_{steam}: Saturated steam temperature (˚C).
   - T_{res}: Initial reservoir temperature (˚C).

2. **Energy Absorption by the Reservoir (Q_{absorbed})**:

$$Q_{\text{absorbed}} = V_{\text{zone}} \cdot (\rho C)_{\text{composite}} \cdot \Delta T$$

$$(\rho C)_{\text{composite}} = (1 - \phi)\rho_r C_{pr} + \phi\left(S_o\rho_o C_{po} + S_w\rho_w C_{pw}\right)$$

   - V_{zone}: Volume of the heated drainage area (m^3).
   - ϕ: Reservoir rock porosity.
   - ρ_r, ρ_o, ρ_w: Densities of the rock matrix, crude oil, and formation water (kg/m^3).
   - C_{pr}, C_{po}, C_{pw}: Specific heat capacities of rock, oil, and water (J/kg ·° C).
   - S_o, S_w: Oil and water saturation fractions.
   - ΔT: Temperature increase, $T_{\text{steam}} - T_{\text{res}}$ (˚C).

Stage 2: Reservoir Thermal Decay (Production / Puff Phase)

Physical Principle

After steaming stops, production begins. Near-wellbore temperature drops because warm fluids are pumped out and heat conducts away into cold surrounding rocks.

1. The Physical Intuition (Newton's Law of Cooling)

After steam injection stops, the near-wellbore region is at peak temperature T_{steam} (e.g., $180°\text{C}$), while the surrounding deep reservoir rock sits at its cold native baseline T_{res} (around $46°\text{C}$).

Heat naturally flows from the hot zone to the cold zone through conduction into adjacent rocks and convection via produced fluids. The fundamental law governing this heat loss states:

$$\text{Rate of Temperature Drop} \propto \text{Current Temperature Difference}$$

$$\frac{dT(t)}{dt} = -\alpha(T(t) - T_{\text{res}})$$

- $\frac{dT(t)}{dt}$: Rate of change of temperature with respect to time (negative because the well is cooling).
- $(T(t) - T_{\text{res}})$: Thermal driving potential (how much hotter the well is than the surrounding earth). As the well cools, this gap shrinks, so cooling slows down over time.
- α: The thermal decay constant (day^{-1}), determined by fluid production rates, rock thermal conductivity, and reservoir thickness.

2. Step-by-Step Mathematical Derivation

To find the well temperature at any future day t, we solve this first-order ordinary differential equation using **separation of variables**.

Step 2.1: Separate Variables

Group the temperature terms on the left and time terms on the right:

$$\frac{1}{T(t) - T_{\text{res}}}\, dT = -\alpha\, dt$$

Step 2.2: Set Up Definite Integrals with Boundary Conditions

- At day $t = 0$ (end of steam soak / start of production), $T(0) = T_{\text{steam}}$.
- At day t (any arbitrary day during production), the temperature is $T(t)$.

$$\int_{T_{\text{steam}}}^{T(t)} \frac{1}{T - T_{\text{res}}}\, dT = \int_0^t -\alpha\, dt$$

Step 2.3: Evaluate Both Integrals

- **Left-hand side:** $\int \frac{1}{u}\, du = \ln|u|$

$$[\ln(T - T_{\text{res}})]_{T_{\text{steam}}}^{T(t)} = \ln(T(t) - T_{\text{res}}) - \ln(T_{\text{steam}} - T_{\text{res}})$$

Using the logarithm quotient rule $\ln(a) - \ln(b) = \ln\left(\frac{a}{b}\right)$:

$$\ln\left(\frac{T(t) - T_{\text{res}}}{T_{\text{steam}} - T_{\text{res}}}\right)$$

- **Right-hand side:**

$$[-\alpha t]_0^t = -\alpha t - 0 = -\alpha t$$

Equating both sides gives:

$$\ln\left(\frac{T(t) - T_{\text{res}}}{T_{\text{steam}} - T_{\text{res}}}\right) = -\alpha t$$

Step 2.4: Exponentiate to Solve for $\boldsymbol{T(t)}$

Take the exponential (e^x) of both sides to remove the natural logarithm:

$$\frac{T(t) - T_{\text{res}}}{T_{\text{steam}} - T_{\text{res}}} = e^{-\alpha t}$$

Multiply both sides by $(T_{\text{steam}} - T_{\text{res}})$:

$$T(t) - T_{\text{res}} = (T_{\text{steam}} - T_{\text{res}})e^{-\alpha t}$$

Add T_{res} to both sides to arrive at the final governing formula:

$$T(t) = T_{\text{res}} + (T_{\text{steam}} - T_{\text{res}})e^{-\alpha t}$$

3. Physical Behavior of the Final Equation

```
 Temperature T(t)
    ^
180°C ───┐  (t = 0, e^0 = 1)
         │ \
         │  \   Exponential cooling
         │   \
 46°C ───┼────'──────────────────────────► Asymptotic Floor (Native T_res)
         └───────────────────────────────► Time (Days)
```

1. **At Day 0 ($t = 0$):**

$$T(0) = T_{\text{res}} + (T_{\text{steam}} - T_{\text{res}}) \cdot e^0 = T_{\text{res}} + T_{\text{steam}} - T_{\text{res}} = T_{\text{steam}}$$

The well begins at maximum steam temperature (e.g., 180°C).

2. **As Time Passes ($t \to \infty$):**

$$e^{-\alpha t} \to 0 \Longrightarrow T(\infty) = T_{\text{res}} + 0 = T_{\text{res}}$$

The temperature asymptotically levels off at native reservoir temperature (e.g., 46°C), ensuring the model never predicts temperatures below the cold earth's native floor.

3. **Why It Matters for the Digital Twin:** This decay curve directly drives the **Andrade Viscosity Model** ($\ln \mu = A + B/T$). As $T(t)$ drops according to this equation, fluid viscosity spikes, triggering the control engine to automatically slow down the sucker rod pump speed to avoid rod float

Mathematical Formulation

Using a lumped-capacitance heat loss formulation:

$$\frac{dT(t)}{dt} = -\alpha(T(t) - T_{\text{res}})$$

Integrating with the initial condition $T(0) = T_{\text{steam}}$:

$$T(t) = T_{\text{res}} + (T_{\text{steam}} - T_{\text{res}})e^{-\alpha t}$$

- $T(t)$: Formation/wellbore temperature at day t (˚C).
- t: Production time elapsed (days).
- α: Thermal decay coefficient (day^{-1}), capturing both conduction to caprock and fluid advection.

Stage 3: Dynamic Crude Viscosity Model

Physical Principle

As the well cools, heavy crude viscosity increases sharply. This temperature-dependent relationship dictates fluid mobility and downhole drag.

Mathematical Formulation

The Andrade-Reynolds exponential relation:

$$\mu(t) = \exp\left(A + \frac{B}{T(t) + 273.15}\right)$$

- $\mu(t)$: Dynamic viscosity of the oil at time t (Pa · s or cP).
- $T(t)$: Fluid temperature at time t (˚C).
- A, B: Empirical constants obtained by fitting laboratory viscometer data.

Stage 4: Sucker Rod Pump (SRP) Dynamics & Wave Mechanics

Physical Principle

During the downstroke, gravity pulls the rod string downward through viscous oil inside the production tubing. If viscous drag equals or exceeds buoyant weight, the rod floats, causing rod compression, buckling, or severe surface-downhole phase lag.

Mathematical Formulation

1. **Laminar Annular Viscous Drag Force (F_{drag})**:

For a cylindrical rod of diameter D_{rod} moving at velocity v_{rod} inside tubing of inner diameter D_{tubing}:

$$F_{\text{drag}}(t) = \frac{2\pi\mu(t)v_{\text{rod}}L_{\text{rod}}}{\ln\left(\frac{D_{\text{tubing}}}{D_{\text{rod}}}\right)}$$

- L_{rod}: Total length of the sucker rod string (m).
- v_{rod}: Linear velocity of the rod string (m/s), proportional to surface Strokes Per Minute (SPM).
- $D_{\text{tubing}}, D_{\text{rod}}$: Tubing inner diameter and rod outer diameter (m).

2. **Buoyant Weight of the Rod String ($W_{\text{rod, buoyant}}$)**:

$$W_{\text{rod, buoyant}} = (\rho_{\text{steel}} - \rho_{\text{fluid}}) \cdot A_{\text{rod}} \cdot L_{\text{rod}} \cdot g$$

- ρ_{steel}: Steel density (≈ 7850 kg/m^3).
- ρ_{fluid}: Wellbore fluid density (kg/m^3).
- A_{rod}: Cross-sectional area of the rod ($\frac{\pi}{4}D_{\text{rod}}^2$) (m^2).
- g: Acceleration due to gravity (9.81 m/s^2).

3. **1D Damped Wave Equation (Gibbs Equation for Rod Elasticity)**:

$$\frac{\partial^2 u(x,t)}{\partial t^2} = a^2 \frac{\partial^2 u(x,t)}{\partial x^2} - c(t)\frac{\partial u(x,t)}{\partial t}$$

- $u(x,t)$: Elastic axial displacement along the rod at depth x and time t (m).
- $a = \sqrt{\frac{E}{\rho_{\text{steel}}}}$: Acoustic speed of sound in the steel rod (≈ 5000 m/s).
- E: Young's modulus of steel (2.1×10^{11} Pa).
- $c(t)$: Viscous damping factor (s^{-1}), parameterized by dynamic fluid viscosity:

$$c(t) = \frac{2\pi\mu(t)}{\rho_{\text{steel}}A_{\text{rod}}\ln\left(\frac{D_{\text{tubing}}}{D_{\text{rod}}}\right)}$$

Stage 5: Digital Twin Optimization & Control Engine

[Stage 1: Steam Heat Input]

|

▼

[Stage 2: Temperature Decay T(t)]

|

▼

[Stage 3: Fluid Viscosity μ(t)]

|

▼

[Stage 4: SRP Mechanics & Drag]

├── Dynamic SPM Limit (No-Float)

└── Downhole Card Reconstruction

|

▼

[Stage 5: Economic Objective J]

└── Trigger Next Huff/Puff Cycle

1. Dynamic Safe Speed Limit ($\mathbf{SPM_{safe}}$)

To prevent rod float, the downward gravitational force must exceed viscous drag ($W_{\text{rod, buoyant}} > F_{\text{drag}}$):

$$v_{\text{rod, max}}(t) < \frac{(\rho_{\text{steel}} - \rho_{\text{fluid}}) g A_{\text{rod}} \ln\left(\frac{D_{\text{tubing}}}{D_{\text{rod}}}\right)}{2\pi\mu(t)}$$

Expressed in allowable pump motor speed:

$$\text{SPM}_{\text{safe}}(t) \leq K \cdot \frac{(\rho_{\text{steel}} - \rho_{\text{fluid}}) g}{\mu(t)}$$

- K: Mechanical geometric scaling constant combining pump stroke length and diameter ratios.

2. Lifecycle Economic Objective Function

The net profit over an entire steam-production cycle (J) separates initial lump-sum injection costs from continuous production revenues and operating expenses:

$$J = \int_0^{t_{\text{cycle}}} [P_{\text{oil}} q_o(t) - C_{\text{elec}} P_{\text{motor}}(t) - C_{\text{wear}} \sigma_{\text{rod}}(t)]\, dt - C_{\text{steam}} m_s$$

- $P_{\text{oil}} q_o(t)$: Revenue from oil production rate $q_o(t)$ at price P_{oil}.
- $C_{\text{elec}} P_{\text{motor}}(t)$: Electricity cost to operate the pump motor.
- $C_{\text{wear}} \sigma_{\text{rod}}(t)$: Equipment wear/fatigue penalty based on peak rod stress.
- $C_{\text{steam}} m_s$: Upfront cost incurred during the steam injection (Huff) stage.

Cycle Cutoff Trigger: The digital twin terminates the production phase and triggers the next steam injection cycle when marginal profit rate drops below operating threshold:

$$\frac{dJ}{dt} = P_{\text{oil}} q_o(t) - C_{\text{elec}} P_{\text{motor}}(t) - C_{\text{wear}} \sigma_{\text{rod}}(t) \leq 0$$

When the net profit rate ($\frac{dJ}{dt}$) drops to zero or the Steam-Oil Ratio (SOR) exceeds economic viability, the digital twin signals that the current production cycle is complete, prompting the next steam injection cycle.